

MANTHAN SHRIRAM JOSHI

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Education

Northeastern University, Boston MA Master of Science in Cell and Gene Therapies <i>Coursework: Stem Cells and Regeneration, Cell and Gene Therapies, Immunology, Regulatory Landscapes, Advanced Drug Delivery Systems, Bioinformatics, Scientific Communication and Biotech Enterprise</i>	December 2026
Harvard Business School, Boston MA Certificate in Management Essentials	August 2025
Manipal Institute of Technology (MAHE), Manipal India Bachelor of Technology in Biotechnology Minor: Pharmaceutical Biotechnology <i>Four-year accredited engineering program. Coursework: Bioprocess Engineering, Downstream Processing, Bioreaction Engineering, Biochemistry, Genetic Engineering, Microbiology, Bioinformatics</i>	July 2023

Skills

Process & Manufacturing	Aseptic technique; process development, optimization, and yield improvement; assay development (dose-response, cytotoxicity, kill curve design); protocol and SOP authorship; experimental troubleshooting and root-cause investigation; cell expansion, banking, and cryopreservation; scientific documentation and technical reporting.
Cell Culture & Genome Engineering	Aseptic handling of mammalian cell lines (CHO) across 5+ passages at >80% viability; end-to-end CRISPR/Cas9 knock-in and knock-out workflow design; Cas9-RNP assembly with target-specific sgRNAs; CRISPRmax lipofection; drug-selection optimization using CellTiter-Glo 2.0 cytotoxicity assays.
Molecular Biology & Analytical	Bacterial transformation, colony screening, miniprep purification; restriction-enzyme digestion for construct verification; genomic DNA extraction; Phusion PCR; primer design in Benchling; Sanger sequencing and ICE indel analysis; agarose gel electrophoresis; Nanodrop quantification; recombinant protein expression and Ni-NTA purification (IMAC); SDS-PAGE; Western blotting; ELISA.
CGT Domain Knowledge	CAR-T (autologous and allogeneic platforms); iPSC differentiation; CRISPR/Cas9 genome engineering; lentiviral vector handling and titer determination; advanced drug delivery systems; cGMP and CGT regulatory frameworks (coursework).
Commercial & Business	Market research and competitive landscaping; investor deck development; grant proposal support; technical writing for non-specialist audiences; client and stakeholder communication; cross-functional project coordination.
Instrumentation & Software	Bioreactor/fermenter operation; HPLC; gas chromatography; fluorescence microscopy; flow cytometry sample preparation; FlowJo (data analysis); Benchling (primer design, construct mapping); GraphPad Prism (dose-response curve fitting); Microsoft Excel; Microsoft Office; ICE web tool; R (basic).

Professional Experience

Northeastern University, Teaching Assistant, Cell and Gene Therapies, Boston, MA	Jan 2026 – Apr 2026
- Taught graduate-level CGT concepts in office hours and discussion sessions, translating current CAR-T and iPSC-to-T-cell literature for students from mixed scientific backgrounds. - Reviewed student project deliverables and delivered structured written feedback; prepared literature summaries on allogeneic and autologous CAR-T platforms to support classroom case discussions.	
Northeastern University, Graduate Research Project, Boston, MA	Sept 2025 – Dec 2025
- Designed and executed an end-to-end CRISPR/Cas9 HPRT1 knockout and knock-in workflow in CHO cells, from guide RNA design and Cas9-RNP delivery through drug selection and sequence confirmation. - Developed and optimized a 6-thioguanine selection protocol using CellTiter-Glo 2.0 dose-response across 8 serial 3-fold dilutions, selecting 100 μM (~7% viability) to enrich HPRT1-KO cells; confirmed by fluorescence microscopy. - Confirmed editing at the sequence level by Phusion PCR and Sanger sequencing with ICE indel analysis, and verified plasmid constructs by restriction digest.	
Reagene Biosciences Pvt Ltd, Research Intern, Bangalore, India	Aug 2023 – Aug 2025
- Built market research and competitive analysis for a 3D-bioprinted human tissue platform positioned as an alternative to animal testing in preclinical safety work. - Contributed to investor decks and grant proposals used in fundraising, translating the underlying science into a commercial case for non-technical funders. - Identified operational and delivery bottlenecks affecting customer fulfilment and raised process gaps with company leadership. - Maintained and cryopreserved mammalian cell lines at high viability to supply the bioink formulation process for 3D-bioprinted tissue models, and authored SOPs for routine laboratory procedures.	
Tranalab Pvt Ltd, Center for Human Genetics, Undergraduate Research Intern Bachelor's Thesis, Bangalore, India	Jan 2023 – Aug 2023
- Designed and executed a complete expression and purification process for M4, a recombinant membrane protein domain, as a bachelor's thesis project ranked first in the cohort. - Identified solubilization as the yield-limiting step of the process and reformulated the buffer accordingly, increasing soluble protein recovery by roughly 48%. - Ran the upstream and downstream train end to end: expression in <i>E. coli</i> under IPTG induction, cell lysis by probe sonication, inclusion body solubilization, and Ni-NTA affinity chromatography (IMAC) with stepwise wash and elution. - Confirmed product identity by SDS-PAGE, quantified yield by A280, buffer-exchanged by dialysis, and assessed refolding by affinity chromatography to verify retained binding function.	

Leadership and Projects

Vice President, Northeastern University, Graduate Biotech-Bioinfo Association, Boston, MA	Dec 2024 – Jan 2026
Led executive board operations and organizational strategy, managing communications and cross-committee coordination while planning networking events, workshops, and industry panel discussions for member engagement.	
CurcuNano, Chief Executive Officer, BIOT 5219 Biotech Enterprise, Northeastern University	Spring 2026
- Led the venture team as CEO for a curcumin nanoformulation concept, developing the business case and market positioning across a semester-long project. - Owned the funding ask in the final investor pitch, covering raise amount, use of funds, and investment rationale, defended to a faculty panel.	